

The Role of Backbone Polarity on Aggregation and Conduction of Ions in Polymer Electrolytes

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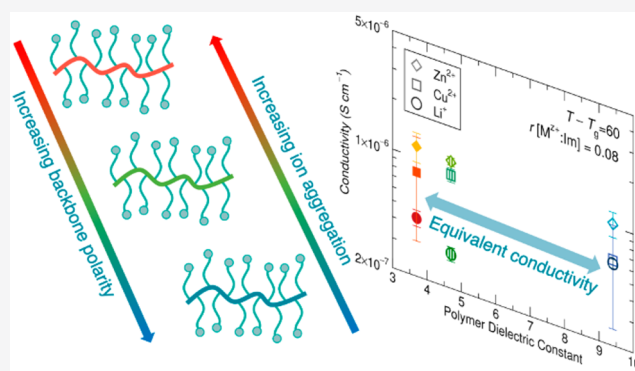


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ABSTRACT: The usual understanding in polymer electrolyte design is that an increase in the polymer dielectric constant results in reduced ion aggregation and therefore increased ionic conductivity. We demonstrate here that in a class of polymers with extensive metal–ligand coordination and tunable dielectric properties, the extent of ionic aggregation is delinked from the ionic conductivity. The polymer systems considered here comprise ether, butadiene, and siloxane backbones with grafted imidazole side-chains, with dissolved Li^+ , Cu^{2+} , or Zn^{2+} salts. The nature of ion aggregation is probed using a combination of X-ray scattering, electron paramagnetic resonance (in the case where the metal cation is Cu^{2+}), and polymer field theory-based simulations. Polymers with less polar backbones (butadiene and siloxane) show stronger ion aggregation in X-ray scattering compared to those with the more polar ether backbone. The T_g -normalized ionic conductivities were however unaffected by the extent of aggregation. The results are explained on the basis of simulations which indicate that polymer backbone polarity does impact the microstructure and the extent of ion aggregation but does not impact percolation, leading to similar ionic conductivity regardless of the extent of ion aggregation. The results emphasize the ability to design for low polymer T_g through backbone modulation, separately from controlling ion-polymer interaction dynamics through ligand choice.



INTRODUCTION

Ion conducting polymers show great potential as electrolytes in energy storage devices because they promise increased safety, mechanical robustness, and voltage stability.^{1,2} Higher ionic conductivity than is currently obtainable is still a requirement for practical applications.³ A fundamental understanding of the design principles for high-conductivity solvent-free polymer electrolytes remains elusive, although significant progress has been made both experimentally^{4–8} and computationally.^{6,9,10}

Ionic conductivity in polymer electrolytes is dependent on the concentration and diffusion coefficient of the ionic species present.¹¹ Many factors influence these quantities, including polymer dielectric constant,^{12–15} glass transition temperature, T_g ,^{16–18} solvation chemistry,^{19–21} ion aggregation,^{22,23} and molecular architecture and morphology (Figure 1).^{24–27} A current challenge is to design systems that enable these factors to be separated in order to identify their respective impacts. This is difficult as many of these factors are interconnected, an example being polymer polarity and T_g .²⁸

The extent of ion aggregation or clustering remains relatively unexplored for many reported polymer electrolytes. The lower dielectric constant of polymer electrolytes compared to organic

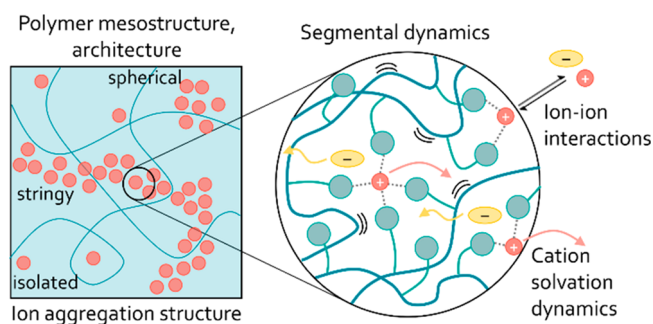


Figure 1. Polymer electrolytes often dissolve salt incompletely due to a low dielectric constant, with ion–ion interactions resulting in aggregation. On a molecular level, the solvation site identity, polymer architecture, segmental dynamics, and salt identity affect ion motion.

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liquid electrolytes results in incomplete salt dissociation and clustering of ionic species added to the polymer. Conflicting literature exists on whether ion clustering and aggregation negatively impact conductivity performance. Some studies have suggested that reducing ion aggregation leads to higher ionic conductivity. For example, for a poly(ethylene oxide) (PEO)-based Li^+ single-ion conductor, a decrease in the polymer dielectric constant with increasing temperature led to increased ion aggregation. This clustering of ions resulted in a low mobile Li^+ content and low conductivity.^{29,30} Ionic conductivity for this PEO-based single-ion conductor was improved upon the addition of a poly(ethylene glycol) plasticizer, due to both a lower T_g and decreased ion aggregation from charge-shielding.³¹ Furthermore, molecular dynamics simulations on polyether electrolytes suggest that increasing their dielectric constant results in decreased ion aggregation and increased ionic conductivity.³²

There are, however, other studies suggesting that aggregation can enhance ionic conductivity. Sinha and Maranas showed that for single-ion conductors based on Li^+ , Cs^+ , or Na^+ , the aggregation state varied with ion identity but the ionic conductivity did not; they attributed this to an invariance of the local dynamics with ion identity as measured via quasi-elastic neutron scattering.³³ Separately, molecular dynamics simulations for single-ion anion conductors suggest that percolation of ion aggregate domains may enhance anion motion by decoupling it from polymer segmental dynamics.³⁴ Additionally, proton conductivity has shown to be dramatically improved in comb polymers that form aggregated nanochannels compared with similar polymers without nanoscale organization.³⁵ Similar trends are seen in precise ionomers forming nanoscale proton transport layers.³⁶ Lastly, a recent study probing the effect of linker polarity in polymeric ionic liquid networks suggested that both ion aggregation and ionic conductivity near T_g are promoted with the use of a lower polarity linker.³⁷

It is very likely that ion aggregate morphology determines the impact of aggregation on ionic conductivity behavior. Ion aggregates that form into discrete clusters, whether spherical, cylindrical, or ramified, might show poor conductivity performance compared to aggregates that form percolated domains through the bulk of the electrolyte.^{23,38} This was suggested to be the cause of the improved conductivity seen in an imidazolium polymer with ions placed along the backbone rather than pendant to the backbone; changing the placement of ion groups within polymers affects the aggregation morphology, with percolated morphologies leading to improved conductivity behavior.^{39–41} Aggregation morphology is challenging to measure experimentally, however, as X-ray scattering and Raman can indicate the existence but not the morphology of aggregates.³⁹ Some groups have obtained real-space aggregate images using electron microscopy.^{42–44} However, aggregate shape is generally identified from computation rather than experiment.^{39,45}

This work analyzes the impact of aggregation on conductivity for imidazole-containing polymers with a range of polymer backbone chemistries and both mono- and multivalent salts. We have employed a versatile synthetic platform using thiol-ene click chemistry to postfunctionalize polymers containing pendant vinyl groups. This enables us to study the effect of segmental dynamics and dielectric constant on ion aggregation and conductivity without changing the solvating species. We find that the backbone identity controls

the dielectric constant and thus the extent of ion aggregation as probed via X-ray scattering and field-theoretic simulations.

Importantly, ion aggregation and salt identity do not correlate with ionic conductivity performance, and instead, only the glass transition temperature plays a major role in determining conductivity. This study shows how polymer electrolyte design can be optimized by controlling segmental dynamics of the polymer matrix through polymer backbone choice, separately from the choice of the pendant solvating unit.

EXPERIMENTAL AND THEORETICAL METHODS

Synthesis of Polymer Backbones. Poly(allyl glycidyl ether) (PAGE) was synthesized via anionic ring-opening polymerization of allyl glycidyl ether using benzyl alcohol deprotonated by potassium naphthalenide as an initiator as previously reported in the literature by Lee et al.⁴⁶ The reaction was performed at 45 °C under argon atmosphere for 24 h. The resulting PAGE was purified by precipitation in hexane and dried under vacuum at 45 °C for 10 h. Poly(vinyl methyl siloxane) (PVMS) was synthesized by anionic polymerization using standard Schlenk line techniques. A 40 mL portion of uninhibited and dry THF was further purified by distillation over *n*-butyl lithium and dried by the addition of 200 μL of *sec*-butyl lithium at 0 °C, after which the solution was allowed to warm to room temperature. The monomer, 1,3,5-trivinyl-1,3,5-trimethyl-cyclotrisiloxane (Gelest), was degassed by three freeze–pump–thaw cycles and used without additional purification. A 200 μL portion of *sec*-butyl lithium was added to THF at 0 °C as initiator, followed by the addition of 3 mL of degassed monomer. The reaction was allowed to proceed for 3 h at 0 °C before termination with degassed methanol. The solution was concentrated and precipitated in methanol three times. Size exclusion chromatography (SEC) was performed on a Waters Alliance HPLC instrument using a refractive index detector and Agilent PLgel 5 μm MiniMIX-D column at 35 °C with THF as the eluent. Dispersity index ($\bar{D}$) was determined against polystyrene calibration standards (Agilent Technologies). The PVMS molecular weight was estimated from SEC, while the PAGE molecular weight was determined using ^1H NMR end-group analysis (Figure S1). Poly(1,2-butadiene) (PBD) with >95% 1,2 addition, $M_n = 9.5$ kDa, and $\bar{D} = 1.04$ was purchased from Polymer Source, Inc. and used without further purification. ^1H NMR spectra were collected on a Varian 600 MHz spectrometer in CDCl_3 or $\text{DMSO}-d_6$.

Synthesis of Imidazole-Functionalized Polymers. *N*-(2-(1*H*-imidazol-1-yl)propyl)-4-mercaptobutanamide (Im-SH) was synthesized as previously reported by Sanoja et al.⁴⁷ The vinyl-containing polymers were fully functionalized by UV (365 nm) activated thiol-ene click chemistry in methanol for PAGE or methanol/THF (20/80) for PBD and PVMS. The resulting polymers (PAGE-Im, PBD-Im, and PVMS-Im) were purified either through dialysis in methanol (SnakeSkin dialysis tubing with a 3.5 kDa MW cutoff, and solvent exchange every 12 h for a total of 5–7 times) or precipitation in acetonitrile, drying *in vacuo* at 55 °C in the presence of phosphorus pentoxide, and immediate transfer to a nitrogen glovebox to mitigate moisture uptake.

Preparation of Metal Salt-Containing Polymers. Polymers were dissolved in anhydrous methanol and drop-cast onto Teflon boats on a hot plate at 60 °C in a nitrogen atmosphere glovebox. Stock solutions (0.05 M) of Li^+ (Alfa Aesar), Cu^{2+} (Solvionic), and Zn^{2+} (Solvionic) bis(trifluoromethylsulfonyl)imide (TFSI^-) salts in anhydrous methanol were added to the Teflon boats in appropriate amounts to reach nominal metal cation-to-imidazole molar ratios, r , of 0.03, 0.08, and 0.12. Nominal molar ratios for the PAGE-Im polymer were slightly lower, at 0.028, 0.073, and 0.11. The samples were dried *in vacuo* at 60 °C overnight in a vacuum chamber in the glovebox (1×10^{-3} Torr) and then at 60 °C for 5 h in a high vacuum oven (4×10^{-8} Torr) to ensure complete removal of solvent. The samples were then transferred into a nitrogen glovebox for storage and measurement. The divalent salts caused uneven gelation and precipitation of

polymer, requiring further homogenization after drying by pressing the samples between Teflon sheets in a hand press at 90 °C.

Thermal Characterization. Standard aluminum DSC pans were loaded with polymer samples in a nitrogen glovebox. The samples were briefly exposed to air during the sealing of the pans and then transferred directly into the DSC. The glass transition temperature (T_g) of each sample was measured using a PerkinElmer DSC 8000 on second heating at 20 °C min⁻¹ at the midpoint of the step transition. The T_g of the PAGE and PVMS before functionalization were measured on a TA Instruments Q2000 DSC on second heating at 20 °C min⁻¹ using the midpoint method. The T_g of PBD before functionalization was taken from literature.⁴⁸ T_g values are shown with error bars of ± 2.5 °C, due to the expected uncertainty of the measurement technique.

Ionic Conductivity Characterization. Total ionic conductivity was measured as a function of temperature on samples sandwiched between parallel ITO blocking electrodes using electrochemical impedance spectroscopy (EIS). The ITO-coated glass electrodes (Thin Film Devices) were cleaned by sonication for 5 min each in detergent, DI water, acetone, and isopropyl alcohol, followed by a 5 min UV/ozone treatment. The electrode thicknesses were measured using a micrometer, after which a double-sided Kapton tape spacer with a 1/8" hole was added to one electrode. Polymer samples were loaded into the hole in the Kapton spacer in a nitrogen glovebox. Samples were heated to about 30 °C above their T_g before being sealed with a second ITO electrode. All samples were then heated to 110 °C and pressed in a hand press. The final stack thickness was measured using a micrometer, and the sample thickness was determined by subtracting the electrode thicknesses. EIS was measured with a Biologic SP-200 potentiostat using a sinusoidal 100 mV signal from 1 MHz to 1 Hz at temperatures ranging from 30 to 110 °C. The data was converted into dielectric storage and loss, and the ionic conductivities were determined from the real component of conductivity at the maximum in $\tan(\delta)$.⁴⁹ Three samples were measured for each composition, with errors reported as standard deviations from the mean.

Dielectric Constant Measurement. Dielectric constant measurements were performed using a Biologic VSP-300 and parallel ITO blocking electrodes. ITO electrodes were cleaned as mentioned above. Conductive copper tape was adhered to the back (glass) side of each electrode, and silver paste was used to make contact between the front ITO electrode and the copper tape. Two 1 mm $\times$ 6 mm double-sided Kapton tape spacers were added to the electrode surface. Dielectric spectra were measured inside a temperature-controlled INSTECH stage with a sinusoidal 100 mV voltage signal from 1 MHz to 100 Hz. For fluid samples (PVMS, PAGE, and PBD), the top ITO electrode was sealed first, and the material was then added into the cell inside a nitrogen glovebox using capillary action. More viscous samples (PAGE-Im, PBD-Im, and PVMS-Im) were spread onto the ITO electrode with Kapton spacers using a spatula, heated on a hot plate inside a nitrogen glovebox to allow the sample to flow, and then sealed with another ITO electrode. The static dielectric constant was extracted from the frequency-dependent spectra at a frequency of 1 MHz and a temperature of -20 °C, where all samples exhibited a phase of around -90° which indicates a purely capacitive response.

X-ray Scattering. Polymer samples were loaded into metal washers in a nitrogen glovebox and covered with Kapton tape to prevent moisture uptake during measurement. X-ray scattering measurements were performed as a function of temperature at the National Synchrotron Light Source II (NSLS-II, beamline 11-BM, Brookhaven National Laboratory), with an X-ray energy of 13.5 keV, and at the Advanced Light Source (ALS, beamline 7.3.3, Lawrence Berkeley National Laboratory), with an X-ray energy of 10 keV. Samples were equilibrated for 15 min at each temperature before collecting exposures. Data processing, including detector distance calibration using a silver behenate standard, reduction of 2D raw SAXS images into 1D intensity versus q curves, and corrections for empty cell scattering were performed using the Nika package for Igor Pro for data taken at the ALS, and using the SciAnalysis software for NSLS-II data.^{50,51}

Electron Paramagnetic Resonance. Pulse EPR experiments at Q-band (34 GHz) were performed on an ELEXYS 580 Bruker spectrometer equipped with a liquid helium cryostat from Oxford Inc. and a 300 W TWT amplifier. Samples were packed into capillaries in a nitrogen glovebox, inserted into 3 mm EPR tubes, and sealed with a cap and parafilm to prevent moisture uptake. The samples were quenched from room temperature to 20 K for the EPR measurements; thus, the structure surrounding the Cu²⁺ is expected to be representative of the polymers at room temperature.

Electron-Spin-Echo (ESE) Detected EPR. The experiments were carried out with the pulse sequence: $\pi/2 - \tau - \pi - \tau - \text{echo}$, mw pulse lengths $t_{\pi/2} = 16$ ns and $t_{\pi} = 32$ ns, and a τ value of 150 ns were used.

Hyperfine Sublevel Correlation (HYSCORE). The experiments were carried out with the pulse sequence $\pi/2 - \tau - \pi/2 - t_1 - \pi - t_2 - \pi/2 - \tau - \text{echo}$. The parameters were: mw pulses of lengths $t_{\pi/2} = t_{\pi} = 16$ ns, starting times 40 ns for t_1 and t_2 and incremented by 6 ns. Spectra with a τ value of 150 ns were recorded at 20 K. A shot repetition rate of 1.25 kHz was used, and a 16-step phase cycle was used to remove unwanted echoes. The time traces of the HYSCORE spectra were baseline corrected with a third-order polynomial, apodized with a Hamming window, and zero filled. After two-dimensional Fourier transformation, the absolute value spectra were calculated. EPR and HYSCORE spectra were simulated using the MATLAB Easyspin package using the "saffron" function, with details provided in the SI.⁵²

Field-Theoretic Simulations. Our molecular model for the metal salt-coordinating polymer is constructed out of two species of monomeric beads (denoted A and B) and two small-molecule salt ion species (denoted + and -). The A- and B-type beads have equal statistical segment lengths b and are linked into continuous Gaussian chains with a backbone of species A having degree of polymerization $N_{bb} = 100$, with $n_{sc} = 19$ equally spaced symmetric diblock side-chains each with lengths $N_{sc} = 2$, and with the B block at the end of the side chain. The charge and bead densities for all species are Gaussian-distributed with a characteristic radius of $a = b/\sqrt{6}$. The short end B-block of the side chains models the pendant imidazole, as indicated in the schematic of this molecular model presented in Figure 2. The

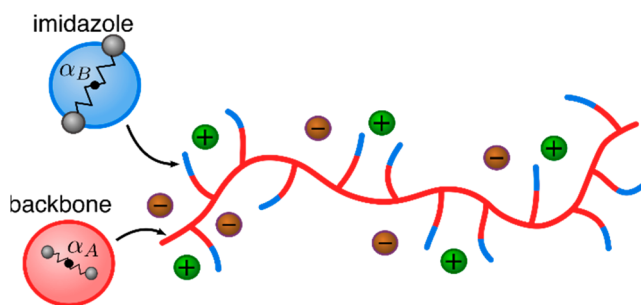


Figure 2. Schematic of the molecular model for the metal salt-coordinating polymer. The backbone monomeric species is shown as red, and the imidazole side chain block is shown as blue, with polarizability volumes α_A and α_B , respectively.

dielectric properties of the polymer are incorporated using embedded Drude oscillators; this polarizable-bead approach is amenable to the transformation to a statistical equilibrium field theory as outlined in previous work by some of us.^{53–56} The backbone monomer is granted polarizability volume $\alpha_v^{(A)}$, which is a parameter that we adjust in this study to control the backbone dielectric constant. For monomeric species B (the imidazole), we set $\alpha_v^{(B)} = 0.541b^3$; we provide justification for this parameter choice in the Supporting Information. For simplicity, we restrict ourselves to monovalent, unpolarizable salt ions in this work and fix the background (vacuum) Bjerrum length to $l_b^{(0)} = 32.5b$ and the average density of the system to $\rho_0 = 7.35b^{-3}$, following previous work.⁵⁶ We model the metal–ligand interaction phenomenologically using a Flory interaction parameter $\chi_{B+} = -20$

between species B and the cation, a common approach in models of salt-polymer complexation.^{17,57,58} We also include short-range repulsions for all beads via an excluded volume parameter $\beta u_0 = \rho_0^{-1} = 0.136b^3$ which sets a weak system compressibility, and Flory parameters $\chi_{AB} = \chi_{+-} = 0.2$ between species A and B, and between the cations and anions (in addition to full electrostatic interactions). Further details on the model and parameter choices are discussed in the SI. Using standard techniques, the field-theoretic canonical partition function takes the form

$$Z = Z_0 \int D\{\omega\} D\varphi e^{-H[\{\omega\}, \varphi]}$$

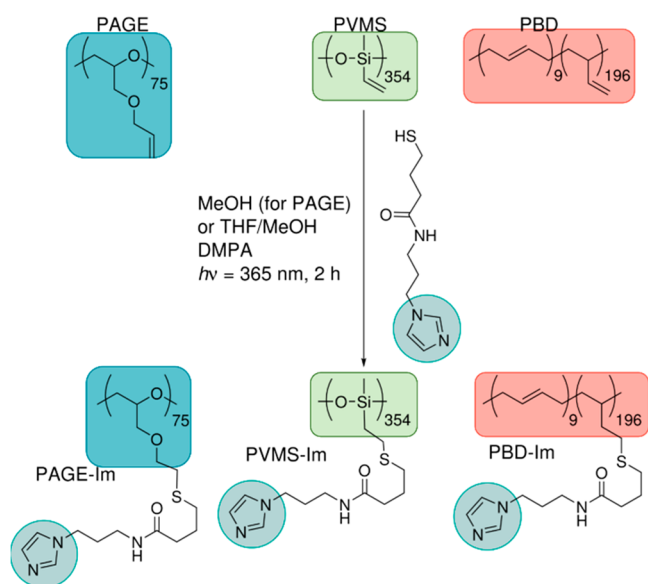
where the $\{\omega\}$ is a set of fluctuating chemical potential and pressure fields that mediate the contact (nonelectrostatic) interactions, and φ is a fluctuating electrostatic potential field, which mediates the ion–ion, ion–dipole, and dipole–dipole interactions in the system.

Traditional simplifying (e.g., mean-field) approximations to polymer field theories are typically not appropriate for charged systems due to the importance of electrostatic field fluctuations and charge correlations. Our interest in ion aggregation in this work makes it particularly critical to capture the effect of charge correlations. The complex Langevin sampling technique allows us to perform field-theoretic simulations (FTS) of the model described above without any approximations. The details of the approach are well described in the literature,^{59–62} and we provide additional specifics regarding our simulations in the Supporting Information.

RESULTS AND DISCUSSION

The goal of this study is to ascertain the impact of polymer backbone polarity on ionic conductivities of both mono and multivalent metal ions. A modular synthetic strategy was employed based on thiol-ene click chemistry which allows the polymer backbone to be altered while keeping the solvating unit, in this case, imidazole, constant for all polymers (see Scheme 1). This requires a set of polymer backbones which have pendant vinyl functional handles. Thus, PAGE, 1,2-PBD, and PVMS were chosen, as they all have vinyl pendant groups on every monomer. These backbones have varying dielectric constants and T_g 's (see Table 1 and Figures S13–S22), both of

Scheme 1. Synthesis of Imidazole-Functionalized Polymers Proceeds via UV-Activated Thiol-ene Click Chemistry in Methanol for PAGE or Methanol/THF (20/80) Mixtures for PVMS and PBD



which are expected to affect salt dissociation and ionic conductivity. The larger molecular weight of the PVMS polymer compared to the PAGE or PBD is not expected to impact the results, since polymer properties such as conductivity and T_g plateau at high molecular weights.⁶³ PAGE has the highest dielectric constant, at 6.0, while PBD has the lowest dielectric constant at 2.3. Functionalization with the imidazole ligand increases both the dielectric constant and T_g of each polymer, with the largest effect on T_g for the siloxane backbone. We also note that the imidazole-thiol chosen for functionalization incorporates an amide group, which likely plays a role in modifying the dielectric environment and interacting with cations. However, since this functional group combination is present in all polymer samples, its effect is intrinsically included in the experimental measurements.

Salt addition increases T_g for all polymer systems as shown in Figure 3. Small amounts of salt result in similar T_g changes for both the monovalent LiTFSI as well as divalent $\text{Zn}(\text{TFSI})_2$ and $\text{Cu}(\text{TFSI})_2$, while at larger salt concentrations, the T_g for the divalent salts increases more significantly than for the monovalent one. This is likely due to the dynamic cross-linking which results upon metal coordination within these polymers.⁴⁷ The similar behavior for all three backbone identities suggests metal–ligand interactions between the pendant imidazole and the salt dominate the T_g trend.⁶⁴

While the T_g behavior is similar for all three polymers, X-ray scattering analysis shows varying amounts of ion clustering or aggregation dependent on the polymer backbone polarity. X-ray scattering profiles taken at 25 °C for a narrow q -range are shown in Figure 4, with the full scattering profiles shown in Figures S23–S25. All three polymer precursors show no correlation peak in the q -range of interest; upon addition of varying concentrations of metal salts, a correlation peak appears for some polymers around $0.15\text{--}0.3 \text{ \AA}^{-1}$ ($4\text{--}2 \text{ nm}$). These distances are on the same order of magnitude for separation expected for evenly distributed cations/anions at the low salt concentrations probed in this study (see Table S1). However, since the peaks do not shift to higher q (smaller distances) with increasing salt concentration, it is likely that the correlation peak corresponds to spacings between domains of high salt density (i.e., ion aggregation), rather than distances between individual ions. Further evidence for this assignment will be provided with the EPR analysis below. For the PAGE-Im and PVMS-Im polymers, this correlation peak is only visible at a higher salt concentration of $r = 0.12$, while the PBD-Im polymer exhibits ion aggregation even at the lowest salt concentration of $r = 0.03$.

The intensity of the peaks changes as a function of both backbone identity and metal salt identity. Samples containing $\text{Zn}(\text{TFSI})_2$ have the strongest ion aggregation peaks, followed by $\text{Cu}(\text{TFSI})_2$. X-ray scattering shows correlation lengths due to differences in electron contrast but does not provide information as to the identity of the species providing that contrast. Thus, it is unclear whether the weaker scattering seen for polymers containing LiTFSI is due to the monovalent nature of Li^+ or the much weaker scattering power of Li^+ compared to those of Zn^{2+} and Cu^{2+} . Nevertheless, clear signatures of aggregation in the LiTFSI-containing PBD-Im polymer are observed at a Li^+ to imidazole ratio of $r = 0.12$. It is expected that the lower polarity of the PBD backbone in PBD-Im results in the enhanced ion aggregation in this polymer compared with those of the PVMS or PAGE systems.

Table 1. Properties of the Polymers in This Study

Polymers	Backbone M_n (kDa)	Backbone $\bar{D}$	Backbone dielectric constant ^d	Functionalized dielectric constant ^d	Backbone T_g (°C) ^e	Functionalized T_g (°C) ^e
PAGE-Im	8.1 ^a	1.17 ^c	6.0	9.4	-73 ± 2.5	-15 ± 2.5
PVMS-Im	29.5 ^b	1.23 ^c	2.9	4.7	-115 ± 2.5	-8 ± 2.5
PBD-Im	9.5	1.04	2.3	3.7	-20 ± 2.5 ⁴⁸	9 ± 2.5

^a M_n measured using NMR end-group analysis. ^b M_n measured via GPC with PS standard. ^c $\bar{D}$ measured using GPC with PS standard. ^dDielectric constants measured at -20 °C at a frequency of 1 MHz. ^e T_g measured using DSC upon second heating.

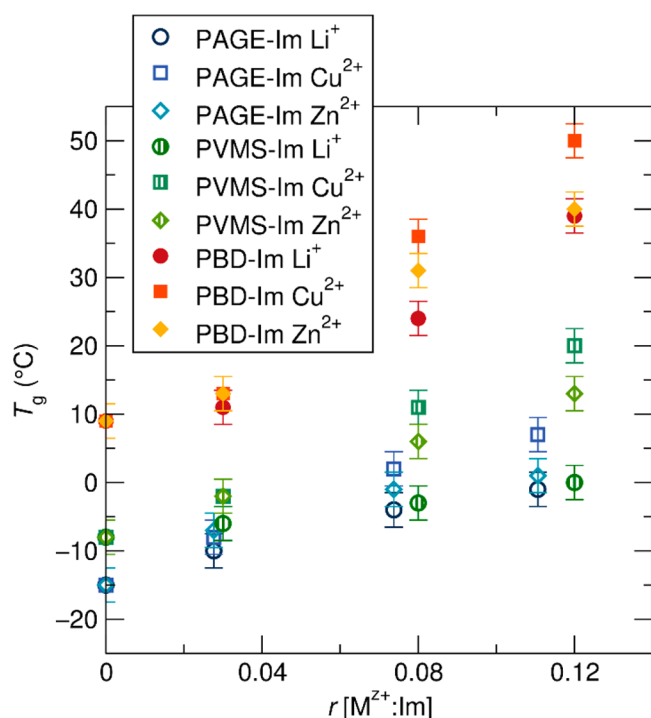


Figure 3. Electrolyte glass transition temperature, T_g (°C) as a function of salt concentration given in molar ratio of metal cation to imidazole ligand (r) for the three polymer systems studied. T_g increases with salt concentration due to dynamic cross-linking interactions between the polymer and the cations. Error bars show uncertainty of ± 2.5 °C.

The effect of temperature on ion aggregation was also probed, showing a slight decrease in peak intensity with increasing temperature as can be seen in Figures S26–S35. This suggests the aggregates become slightly more diffuse as temperature increases but remain present through the entire temperature range of interest.

Surprisingly, while the three polymers show dramatic differences in ion aggregation as probed via X-ray scattering, these effects are not manifested in the ionic conductivity behavior. Figure 5 shows the ionic conductivity plotted as a function of salt concentration for all the polymers studied, at a temperature 60 °C above the glass transition temperature for each system. This representation removes effects of changes in T_g with polymer identity, salt identity, or salt concentration which were discussed previously (see Figure 3). A temperature of 60 °C above T_g was chosen because this temperature ranged from 49 to 110 °C for the various samples, which was directly measurable. Figures S10–S11 show the same conductivity trend at relative temperatures of $T - T_g = 30$ and $T - T_g = 90$. Ion aggregation was visible in the X-ray scattering data throughout this temperature range (Figures S26–S35). No clear trend in conductivity exists as a function of polymer

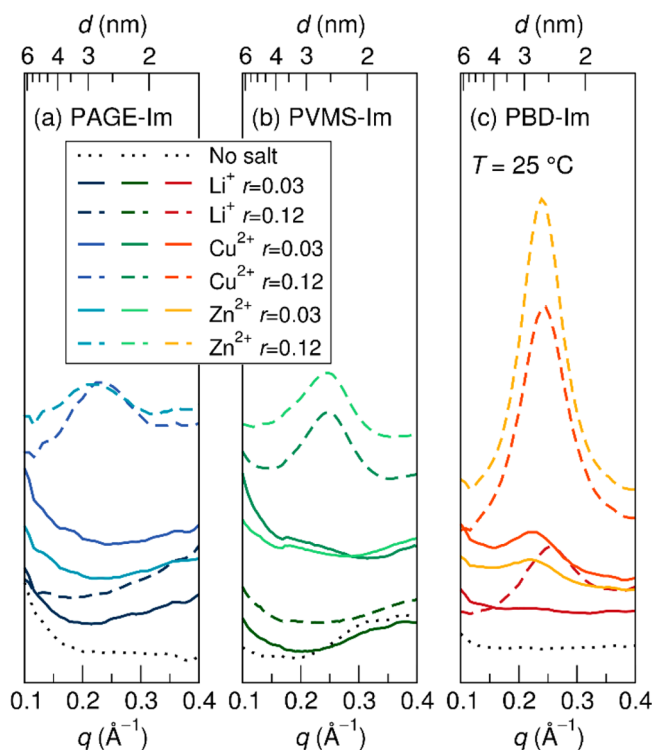


Figure 4. X-ray scattering profiles for the (a) PAGE-Im, (b) PVMS-Im, and (c) PBD-Im polymers as a function of salt identity and concentration. Peaks in the scattering profile indicate ion aggregation.

backbone or salt identity. Since the PAGE-Im backbone exhibited the least ion aggregation, especially for the LiTFSI salt, one might have expected the highest conductivities for this system. However, the PBD-Im and PVMS-Im polymers show comparable conductivities for all salts at all concentrations. This suggests that ion aggregation, as measured via X-ray scattering, does not significantly affect conductivity behavior. Instead, as Figure S12, which plots ionic conductivity at an absolute temperature of 60 °C for each system, shows, the glass transition temperature is the strongest indicator for conductivity performance in these systems.

Understanding how the backbone identity affects the ion aggregate microstructure and ion–ion correlations without significantly impacting ionic conductivity requires us to investigate the nature of the microstructure and correlations in these systems. To this end, we turn to electron paramagnetic resonance (EPR) to probe local ion environments and field-theoretic simulations to identify ion aggregation morphology and characteristics.

While X-ray scattering shows bulk changes in ion aggregation as a function of polymer identity, EPR can be used to identify local changes in the environment of paramagnetic metals. Here, we exploited the EPR properties

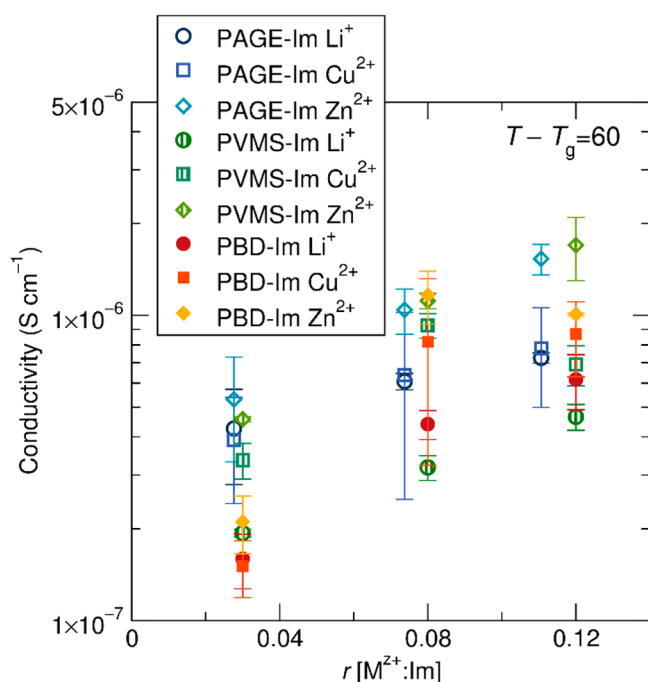


Figure 5. Ionic conductivity is relatively invariant with polymer backbone and salt identity when normalized by the T_g of each system, suggesting ion aggregation plays a minimal role in affecting conductivity.

of Cu^{2+} with an electron configuration of d^9 , and hence, one unpaired electron in its d-orbital. The copper nucleus itself has two magnetically active isotopes (^{63}Cu 69.13%; ^{65}Cu 30.83%) with nuclear spin $I = 3/2$, thus resulting in hyperfine interactions between the electron and the nuclear spins. The strong spin–orbit coupling and electron–ligand field interactions give rise to significant anisotropy of the g and hyperfine values, both of which are sensitive to the local environment of the Cu^{2+} . Thus, Cu^{2+} was used as a local probe to identify whether the environment around the metal atom is affected by the polymer backbone. Since the largest changes in ion aggregation were observed for the PAGE-Im and PBD-Im polymers, these two systems were chosen for the EPR study. A concentration of $r = 0.03$ was used to probe systems where the ion aggregation as probed via X-ray scattering differed between the two polymers, with the PAGE-Im polymer showing no ion aggregation but the PBD-Im polymer exhibiting signs of ion aggregation.

Field-Swept Echo (FSE) spectra acquired for PAGE-Im Cu^{2+} and PBD-Im Cu^{2+} at 20 K, shown in Figure 6a, indicate the presence of local ion–ion interactions in both polymers. While the axial Cu^{2+} lineshapes for each of the backbones show subtle differences, both display significant broadening contributions. The spectra show an envelope of the different orientations of the Cu^{2+} spins. Simulation of the line shape allows us to model the differences in interactions that result from differences in structural nuances between the two backbones (Figure 6b). The EPR parameters extracted from the simulations are summarized in Table 2. The lineshapes for the PBD-Im and PAGE-Im can only be simulated considering a two-component system of (i) resolved, isolated, Cu^{2+} ions with an axially elongated tetragonal ligand field ($g_{\parallel} > g_{\perp}$) and (ii) a broad, unresolved component, corresponding to clusters of ions, hinting toward ion aggregation even at such low

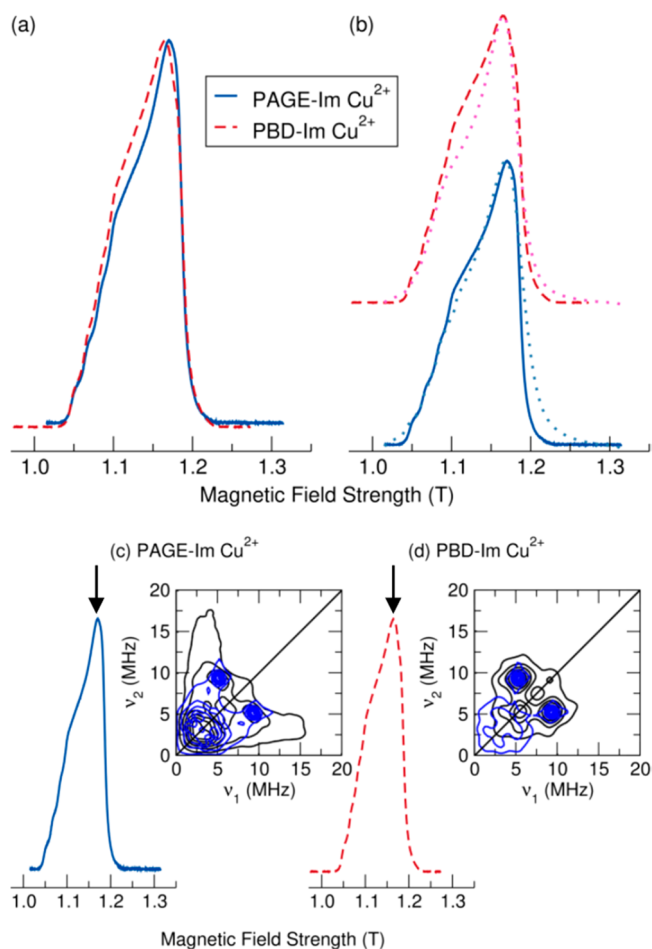


Figure 6. (a) FSE of the two polymers with different backbones, recorded at 20 K. (b) Comparison of experimental (solid/dashed lines) and simulated spectra (dotted lines) for each polymer. PBD spectra are shifted vertically for clarity. Experimental (black) and simulated (blue) HSCORE spectra of (c) Cu-PAGE-Im and (d) Cu-PBD-Im at the maximum observer field position at 20 K.

Table 2. Experimentally Derived g and A Parameters and Contributions from Narrow and Broad Components for Both Polymer Backbones

	$g_{\parallel}$ (± 0.005)	$g_{\perp}$ (± 0.005)	$A_{\parallel}(\pm 10)$ (MHz)	$A_{\perp}(\pm 10)$ (MHz)	%
PAGE-Im Cu^{2+} (I)	2.245	2.045	616	112	25
PAGE-Im Cu^{2+} (II)	2.245	2.045			75
PBD-Im Cu^{2+} (I)	2.250	2.050	686	224	25
PBD-Im Cu^{2+} (II)	2.250	2.050			75

concentrations (Figure S36). The broad component accounts for a significant proportion of EPR active species within the spectrum ($\sim 75\%$) for both polymers. Dipolar coupling is expected to occur in EPR when the Cu^{2+} are within about 1.5 nm from each other.^{65,66} A minimum Cu^{2+} concentration for dipolar interaction without aggregation can be calculated, assuming dipolar broadening occurs at separation distances below 15 Å and between evenly spaced random close packed Cu^{2+} ions. This corresponds to a Cu^{2+} density of around one Cu^{2+} per 170 Å³ (see SI for calculation). Meanwhile, a molar

ratio of $r = 0.03$ as probed here corresponds to a Cu^{2+} concentration of around one Cu^{2+} per 2 nm^3 (see SI for calculation). The contribution of 75% EPR active species to a broadened signal is significantly higher than would be expected from a random distribution of Cu^{2+} ions within the polymer, suggesting that the broad signal indeed comes from aggregated Cu^{2+} . While we are unable to see ion aggregation peaks for such concentrations of Cu^{2+} on the X-ray scattering profiles for the PAGE backbone, X-ray scattering only detects correlations between aggregate clusters rather than between individual ions within an aggregate. In contrast, EPR provides a more sensitive local probe suggesting that multiple Cu^{2+} ions are in close proximity in both backbones, regardless of whether larger scale clustering of ion aggregates occurs or not. Experimentally derived $g_{\parallel}$ and $g_{\perp}$ values listed in Table 2 are typical of Cu^{2+} -imidazole systems, hinting toward binding of the Cu^{2+} and imidazole ligand in both polymers. Strikingly, the hyperfine interactions, both $A_{\parallel}$ and $A_{\perp}$, between the unpaired electron and the copper nuclei are larger than generally observed values for Cu^{2+} -imidazole systems ($A_{\parallel} = 426$ to 615 MHz and $A_{\perp} = 39$ to 50 MHz).⁶⁷ The stronger hyperfine interactions suggest greater elongation along both the z -axis and the x - y plane, possibly due to increased steric hindrance from the imidazole tethered to a polymer backbone. The PBD backbone in fact shows even greater hyperfine coupling for both hyperfine components, indicating that the identity of the backbone alters the local environment of the Cu^{2+} metal ion itself. This can be correlated with the backbone's lower polarity, consequently resulting in greater distortion of the environment of the Cu^{2+} , leading to the observance of the ion aggregation peak in the X-ray scattering profile.

Lineshape analysis of the Cu^{2+} spectra revealed signatures of the local environment of Cu^{2+} . However, to probe the interactions between the Cu^{2+} center with the polymer side-chains and the effect of the backbone, we can rely on resolving hyperfine interactions to the ^{14}N nuclei. The question is whether there is differential interaction between Cu^{2+} and ^{14}N when the backbone is altered. In the field-swept echo (FSE) spectra for both polymers, the hyperfine coupling between Cu^{2+} and ^{14}N is hidden within the larger hyperfine interactions between the unpaired electron spins of Cu^{2+} and the ^{63}Cu or ^{65}Cu nuclear spins and hence not resolved. To closely inspect differences in interaction between Cu^{2+} and imidazole, we used two-dimensional hyperfine sublevel correlation (HYSCORE) spectroscopy, a well-established technique that provides correlations between nuclear frequencies of different spin manifolds that interact with the unpaired electron.⁶⁸ The representative HYSCORE spectra collected for the two backbone polymers (Figure 6c,d) show direct interaction of Cu^{2+} and ^{14}N on the imidazole ligand, as expected. The presence of characteristic dq cross-peaks at approximately $[5.05, 9.10] \text{ MHz}$ and $[9.10, 5.05] \text{ MHz}$ in the strong coupling regime ($A > \nu_{^{14}\text{N}}/2$) shows direct coordination of ^{14}N species to the unpaired electrons in Cu^{2+} in both polymer backbones. In addition, a broad anisotropic peak appears in the lower frequency regime centered around $[3.6, 3.6] \text{ MHz}$ for PAGE-Im Cu^{2+} , but not for PDB-Im Cu^{2+} .

Simulations of the HYSCORE data and estimation of the ^{14}N parameters suggest slight differences in Cu^{2+} -imidazole environments between the two backbones. While a single component was successfully fit to the PBD-Im spectra, a second ^{14}N component was required to produce a reasonable fit for the PAGE-Im Cu^{2+} system (Table S2). This second

component still corresponds to an imidazole ligand (see SI) but shows slightly less electron delocalization and slightly higher symmetry. Overall, for the PAGE backbone, the EPR analysis suggests that the coordination environment of imidazole bound ligands is more heterogeneous due to the presence of different ^{14}N imidazole species, which coordinate to the Cu^{2+} center differently. We cannot exclude the possibility of an interaction between the Cu^{2+} with the ether oxygens of the backbone in this polymer. ^{16}O has a nuclear spin $I = 0$ and therefore cannot be detected by hyperfine techniques. This would result in a decreased interaction with the imidazole ligand. It should also be noted that cross-suppression of peaks is an effect in HYSCORE, where strongly coupled nuclei can suppress weaker couplings with other nuclei.⁶⁹ In other words, interactions with weakly coupled ^{14}N in the amide linkages of the side chains may be suppressed due to stronger interactions between Cu^{2+} and the imidazole.

The EPR results suggest both polymer backbones exhibit local Cu^{2+} - Cu^{2+} interactions, but differ in their Cu^{2+} -imidazole interactions. The general similarity in local ion-ion interactions is perhaps surprising given the difference observed in extent of aggregation according to X-ray scattering and in backbone polarity. However, both polymers have generally low dielectric constants and could be expected to promote ion clustering at a local level, as detected by FSE EPR. The HYSCORE data indicates some inhomogeneity of Cu^{2+} -imidazole interactions in the two backbone systems; changes in solvation behavior, which may impact ion screening, might result in more diffuse ion aggregation which is not detected in X-ray scattering because of the breadth of any resulting peak. Importantly, while EPR confirms the presence of ion aggregates in both polymers, it gives no indication of the morphology or connectivity of the ion aggregates, which likely also plays a large role in ionic conductivity behavior.

Field theoretic simulations examine not only the ion-ion correlations but also visualize the ion densities and thus infer structural information directly. Within our statistical field theory approach, we cannot probe instantaneous density configurations, since instantaneous density field operators are complex-valued. Instead, structures are extracted from time-averages over a finite sampling interval for which the averages of the imaginary parts of the field operators have diminished. As discussed in the SI, the choice of sampling interval is a balance between inaccurate structures at too small averaging windows and total loss of structure for too large windows as the full configuration space for the system (including the ion aggregates) is explored.

An examination of the ion-ion correlations in field-theoretic simulations of the model reveals the presence of a correlation peak, a well-known signature of ion aggregation, consistent with both the X-ray scattering and EPR results above. Figure 7 shows the cation-cation structure factor from FTS for a range of salt-loading and for low and high backbone dielectric constants (referred to as PBD-Im-like and PAGE-Im-like polymers). A peak in the ion-ion structure factor emerges for high salt-loading ($r > 0.026$), significantly stronger for a low backbone dielectric constant ($\epsilon = 2$) than for a high dielectric constant ($\epsilon = 10$). No peak is observed for the low salt-loading cases, in agreement with the trends seen via X-ray scattering of the lithium salt-doped systems. The peak in FTS occurs at a wavevector $q^* \approx 0.5 \text{ \AA}^{-1}$, which may be converted to units of $\text{\AA}^{-1}$ by estimating the statistical segment length $b \approx 0.5 \text{ nm}$ to give $q^* \approx 0.25 \text{ \AA}^{-1}$, in good agreement with the positions of

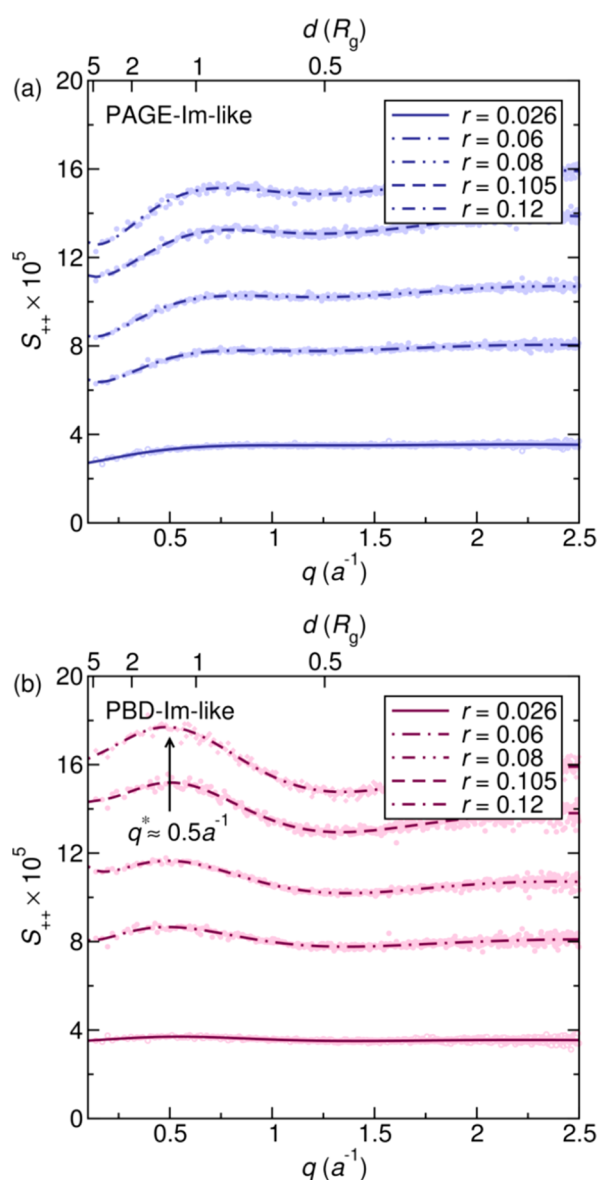


Figure 7. Cation–cation structure factor profiles from simulations of the field-theoretic model for (a) PAGE-Im-like and (b) PBD-Im-like polymers for a range of salt concentrations ($r = 0.026$ to 0.12). The points are raw data from field-theoretic simulations, and the lines are visual guides generated via a nonlinear curve fit of the raw data.

the experimentally observed scattering peaks (Figure 8). However, the peak is significantly broader than that seen in the X-ray scattering; we attribute this to the coarse-grained nature of our model, which produces weaker ion–ion correlations due to the soft (Gaussian) nature of the ions.

The cation–cation structure factors in Figure 7 establish that the field-theoretic model exhibits ion correlations that are consistent with the X-ray scattering results, as shown more clearly in Figure 8: the dependence of structure factors on backbone dielectric constant and salt concentration are in qualitative agreement, and the location of the correlation peak is in quantitative agreement for the PBD-Im polymer. The temporally averaged cation densities from field-theoretic simulations are shown in Figure 9a for $r = 0.105$, for which the structure factor profiles in Figure 7 indicate significant differences in the extent of ion aggregation for the PAGE-Im-like and PBD-Im-like polymers. Visually, the difference

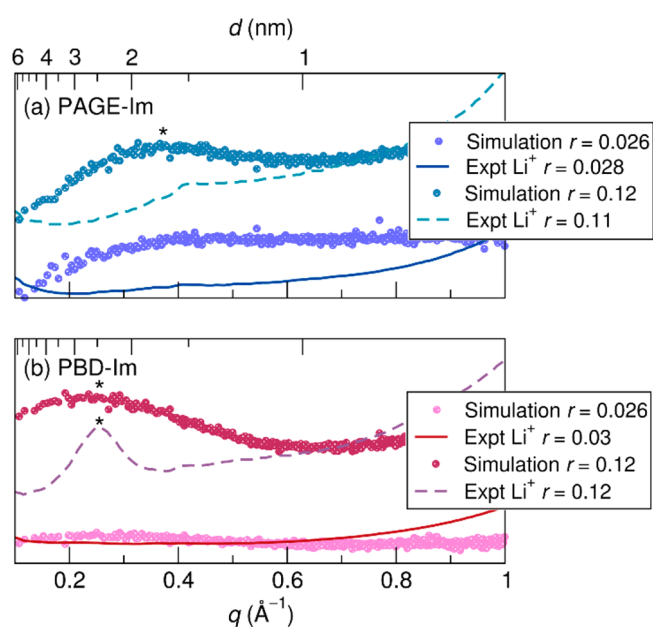


Figure 8. Comparison of experimental X-ray scattering data with FTS cation–cation structure factors for (a) the PAGE-Im polymer and (b) the PBD-Im polymer mixed with LiTFSI. Experimental and simulation data have been scaled vertically for comparison. Peak maxima are indicated by a *.

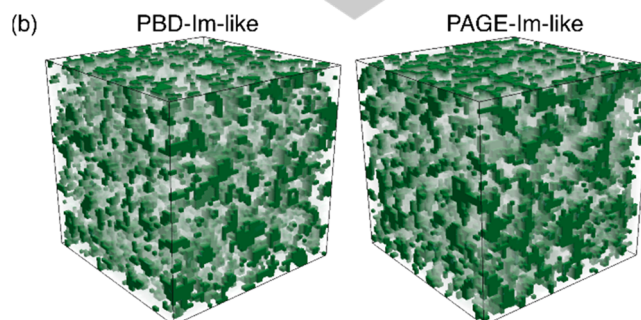
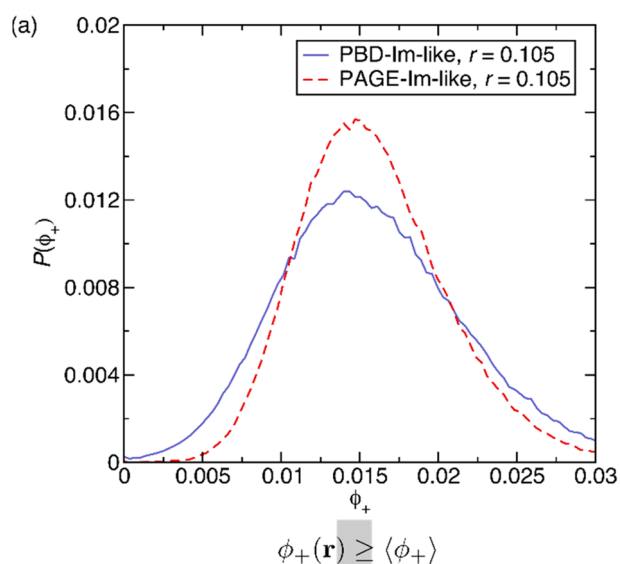


Figure 9. (a) Histograms of the thermally averaged cation density from FTS for the $r = 0.105$ cases. (b) Ion channels, identified by the criterion $\phi_+(r) \geq \langle \phi_+ \rangle$, form a percolating network for both polymers.

between the cation densities for those cases does not appear to be so great, but the histograms of the cation density reveal a measurable difference. Although for strongly segregated ion channels a bimodal ion density distribution is expected, due to high ion concentration in the ion channels and concomitant low ion concentration in the surrounding matrix, in Figure 9a, the cation density is distributed *unimodally* for both cases. This indicates that when averaging over the rearranging ion aggregates, the resulting ion structure can be understood as fluctuations about the bulk ion density, with a characteristic length-scale and a magnitude that depends on the length of the averaging window used. There is a dependence of the magnitude of these fluctuations on the polymer identity (i.e., backbone polarity); that is, the magnitude of the fluctuations is stronger for the PBD-Im-like (low polarity) polymer than for the PAGE-Im-like (high polarity) polymer, as can be seen in the width of the cation density distributions of Figure 9a. This is a manifestation of the same ion aggregation effects that produce the difference in the structure factors in Figure 7.

A rudimentary percolation analysis of the averaged ion morphologies reveals that any differences in the magnitude of the ion density in these channels, due to the difference in polymer backbone polarity, is not sufficient to lead to a substantial difference in their percolation properties. We identify “ion channels” as those for which the cation volume fraction $\phi_+(r)$ is greater than its bulk value (in this case, $r = 0.105$ corresponds to $\langle\phi_+\rangle \approx 0.014$). The resulting channels, which are visualized in Figure 9b, are analyzed to determine whether or not they percolate through the simulation box. In all directions, the channels, as defined by the threshold value $\phi_+ \approx 0.014$, form a completely percolating network; that is, the ratio of the volume of percolating ion channels to the total volume of ion channels is greater than 0.99. Since the ionic conductivity stems from both cation and anion contributions, it is relevant to confirm that the anion–anion structure factor likewise shows behavior similar to the cation–cation structure factor. Indeed, as can be seen in Figure S40, the anion structure factor and percolation behavior are the same as those for the cation.

These results indicate that it is possible for changes in backbone dielectric properties to affect ion aggregation, in a way that would produce differences in the scattering profiles but would *not* produce differences in the percolation properties of the ion channels/network, and thus might not correlate with the ionic conductivity, as was seen in Figure 5.

CONCLUSION

Ionic conductivity in a series of imidazole-tethered graft polymers has been shown to be insensitive to backbone polarity, despite the extent of ion aggregation being greater in the systems with less polar backbones. EPR results reveal that low concentrations of $\text{Cu}(\text{TFSI})_2$ in the butadiene and ether-based polymers show similar Cu^{2+} – Cu^{2+} local interactions, indicating that local ion–ion interactions may still be prevalent in all samples even when not visible in X-ray scattering. Field-theoretic simulations suggest ionic aggregates in both the low- and high-dielectric backbone systems are percolated, despite a dependence of the extent of ion aggregation on the backbone dielectric constant, further supporting the similar conductivity performance in these systems when normalized by T_g . These results underscore that ion aggregation is not necessarily detrimental to ionic conductivity, especially in systems where such aggregates form percolating domains with high local

mobility. The results emphasize the much larger experimental synthetic space, such as focusing on low T_g and noninteracting backbones, that may be accessible in the search for higher performance ion-conducting polymers, than was previously considered viable.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.0c00587>.

NMR and GPC molecular characterization, temperature-dependent ionic conductivity and X-ray scattering, dielectric constant measurements, details of HYSCORE simulation, and field-theoretic simulations (PDF)

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Notes

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